

Data Loading, Handling, and Preprocessing

1 Backend parsers

Snapshot.from_openmx()
Snapshot.from_fhiaims()
Snapshot.from_pyscf()



OpenMX



FHI-aims

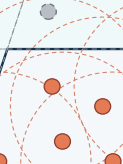
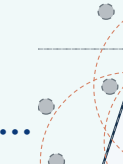
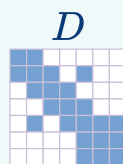
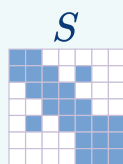
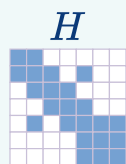


PySCF

...

2 Snapshot

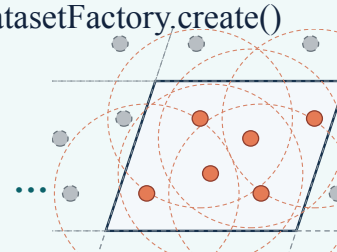
$\{L, R, Z\}$ and $\{H, S, D, E_b, N_e\}$



Lattice, positions, atomic species
Hamiltonian, overlap, density, band energy

3 Dataset

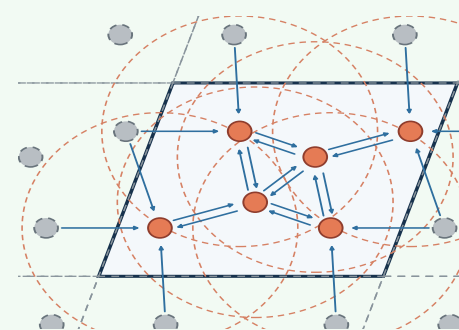
DatasetFactory.add_snapshot()
DatasetFactory.add_snapshot()
DatasetFactory.create()



Factory-driven dataset assembly

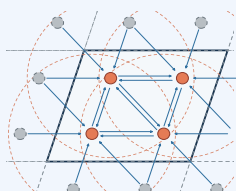
4 Graph sample assembly

compute_graph_features()



Snapshot preparation

- Deterministic, numerical accuracy-optimized edge ordering
- Spherical harmonics convention conversion between backends
- Optional edge filtering
- Observable precomputation



Observables



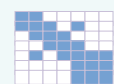
distance



Shared dataset state

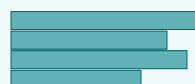
- Global BlockIrrepMapper based on the Wigner-Eckart theorem, holding the Clebsch-Gordan coefficients
- Species-specific orbital irrep-set metadata
- Reusable train / validation / test splits
- Snapshot cache system for accelerated repeat experiments and hyperparameter optimization

Blocks



Block irrep mapper

Irreps



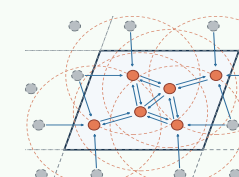
5

GNNDataset sample payload

Input

$$x = \{e_{ij}, Y_{\ell m}(\hat{\mathbf{r}}_{ij}), \phi(|\mathbf{r}_{ij}|), z_i\}$$

Edge list, spherical harmonics,
radial encoding, atomic species



Target

$$y = \{H, S, D, E_b, N_e\}$$

Hamiltonian, overlap, density matrices,
band energy, number of electrons

